

# Common questions about VMware to Azure replication

This article answers common questions that might come up when you deploy disaster recovery of on-premises VMware virtual machines (VMs) to Azure.

## Important

The classic experience to protect VMware machines by using ASR has retired on March 30, 2026. [Learn more](#). Switch to the [modernized experience](#) to avoid service interruption.

## Note

You can now move your existing replicated items to modernized VMware disaster recovery experience. [Learn more](#).

## General

### How do I use the classic experience in the Recovery Services vault rather than the modernized experience?

You can't move to the classic experience in a newly created Recovery Services vault because it will be [deprecated](#) in March 2026. All new Recovery Services vaults use the modernized experience.

### Can I migrate to the modernized experience?

You can migrate all VMware VMs or physical servers that you're replicating by using the classic experience to the modernized experience. Check the details [here](#) and follow the [tutorial](#).

### What do I need for VMware VM disaster recovery?

[Learn about the components involved](#) in disaster recovery of VMware VMs.

## Can I use Site Recovery to migrate VMware VMs to Azure?

Yes. In addition to using Site Recovery to set up full disaster recovery for VMware VMs, you can also use Site Recovery to migrate on-premises VMware VMs to Azure. In this scenario, you replicate on-premises VMware VMs to Azure Storage. Then, you fail over from on-premises to Azure. After failover, your apps and workloads are available and running on Azure VMs. The process is like setting up full disaster recovery, except that in a migration you can't fail back from Azure.

## Does my Azure account need permissions to create VMs?

If you're a subscription administrator, you have the replication permissions you need. If you're not an administrator, you need permissions to take these actions:

- Create an Azure VM in the resource group and virtual network you specify when you configure Site Recovery.
- Write to the selected storage account or managed disk based on your configuration.

[Learn more](#) about required permissions.

## What applications can I replicate?

You can replicate any app or workload running on a VMware VM that meets the [replication requirements](#).

- Site Recovery supports application-aware replication, so that apps can be failed over and failed back to an intelligent state.
- Site Recovery integrates with Microsoft applications such as SharePoint, Exchange, Dynamics, SQL Server, and Active Directory. It also works closely with leading vendors, including Oracle, SAP, IBM, and Red Hat.

[Learn more](#) about workload protection.

## Can I use a guest OS server license on Azure?

Yes, Microsoft Software Assurance customers can use [Azure Hybrid Benefit](#) to save on licensing costs for Windows Server machines that are migrated to Azure, or to use Azure for disaster recovery.

# Security

## What access to VMware servers does Site Recovery need?

Site Recovery needs access to VMware servers to:

- Set up a VMware VM running the Site Recovery configuration server.
- Automatically discover VMs for replication.

## What access to VMware VMs does Site Recovery need?

- To replicate, a VMware VM must have the Site Recovery Mobility service installed and running. You can deploy the tool manually, or you can specify that Site Recovery do a push installation of the service when you enable replication for a VM.
- During replication, VMs communicate with Site Recovery as follows:
  - VMs communicate with the configuration server on HTTPS port 443 for replication management.
  - VMs send replication data to the process server on HTTPS port 9443. You can modify this setting.
  - If you enable mult-VM consistency, VMs communicate with each other over port 20004.

## Is replication data sent to Site Recovery?

No, Site Recovery doesn't intercept replicated data and doesn't have any information about what's running on your VMs. Replication data is exchanged between VMware hypervisors and Azure Storage. Site Recovery has no ability to intercept that data. Only the metadata needed to orchestrate replication and failover is sent to the Site Recovery service.

Site Recovery is certified for ISO 27001:2013 and 27018, HIPAA, and DPA. It's in the process of SOC2 and FedRAMP JAB assessments.

# Pricing

## How do I calculate approximate charges for VMware disaster recovery?

Use the [pricing calculator](#) to estimate costs while using Site Recovery.

For a detailed estimate of costs, run the deployment planner tool for [VMware](#) and use the [cost estimation report](#).

## Is there any difference in cost between replicating to storage or directly to managed disks?

Managed disks have a different pricing structure than storage accounts. [Learn more](#) about managed-disk pricing.

## Is there any difference in cost when replicating to General Purpose v2 storage account?

You typically see an increase in transaction costs on GPv2 storage accounts since Azure Site Recovery is transaction heavy. [Read more](#) to estimate the change.

## Does Site Recovery work with reserved instances?

Yes, you can purchase [reserved Azure virtual machines](#) in the disaster recovery region, and Site Recovery failover operations use them. No extra configuration is needed.

## Mobility service

### Where can I find the Mobility service installers?

Find the installers in the %ProgramData%\ASR\home\svsystems\pushinstallsvc\repository folder on the configuration server.

### How do I install the Mobility service?

On each VM that you want to replicate, install the service by using one of several methods:

- [Push installation](#)
- [Manual installation](#) from the UI or PowerShell
- Deployment by using a deployment tool such as [Configuration Manager](#)

## Managed disks

## Where does Site Recovery replicate data?

Site Recovery replicates on-premises VMware VMs and physical servers to managed disks in Azure.

- The Site Recovery process server writes replication logs to a cache storage account in the target region.
- These logs create recovery points on Azure managed disks that have the prefix **asrseeddisk**.
- When failover occurs, the recovery point you select creates a new target managed disk. This managed disk attaches to the VM in Azure.
- VMs that previously replicated to a storage account (before March 2019) aren't affected.

## Can I replicate new machines to storage accounts?

No. Beginning in March 2019, in the Azure portal, you can replicate only to Azure managed disks.

Replication of new VMs to a storage account is available only by using PowerShell ([Az.RecoveryServices module version 1.4.5](#) ) or the REST API (version 2018-01-10 or 2016-08-10). [Learn how](#) to set up replication using the PowerShell commands.

## What are the benefits of replicating to managed disks?

[Learn how](#) Site Recovery simplifies disaster recovery with managed disks.

## Can I change the managed-disk type after a machine is protected?

Yes, you can easily [change the type of managed disk](#) for ongoing replications. Before changing the type, ensure that no shared access signature URL is generated on the managed disk:

1. Go to the **managed disk** resource on the Azure portal and check whether you have a shared access signature URL banner on the **Overview** blade.
2. If the banner is present, select it to cancel the ongoing export.
3. Change the type of the disk within the next few minutes. If you change the managed-disk type, wait for fresh recovery points to be generated by Azure Site Recovery.
4. Use the new recovery points for any test failover or failover in the future.

## Can I switch replication from managed disks to unmanaged disks?

No. Switching from managed to unmanaged isn't supported.

## Replication

### What are the replicated VM requirements?

[Learn more](#) about support requirements for VMware VMs and physical servers.

### How often can I replicate to Azure?

Replication is continuous when replicating VMware VMs to Azure.

### Can I extend replication?

Extended or chained replication isn't supported. Request this feature in the [feedback forum](#) .

### How can I track progress of initial replication/synchronization?

This capability was recently added to Site Recovery services. Update your Site Recovery infrastructure (configuration servers, scale-out process servers) and mobility agent to versions 9.36 or higher to get accurate details. Learn more on how to track the progress [here](#).

### Can I do an offline initial replication?

Offline replication isn't supported. Request this feature in the [feedback forum](#) .

### What is asrseeddisk?

For every source disk, the service replicates data to a managed disk in Azure. This disk uses the prefix **asrseeddisk**. It stores the copy of the source disk and all the recovery point snapshots.

### Can I exclude disks from replication?

Yes, you can exclude disks.

## **Does upgrading kernel firmware on a server protected by Azure Site Recovery for disaster recovery have any impact?**

No, it doesn't impact ongoing replication because Azure Site Recovery already protects the server.

## **Can I replicate VMs that have dynamic disks?**

You can replicate dynamic disks. The operating system disk must be a basic disk.

## **If I use replication groups for multi-VM consistency, can I add a new VM to an existing replication group?**

Yes, you can add new VMs to an existing replication group when you enable replication for them. However:

- You can't add a VM to an existing replication group after replication starts.
- You can't create a replication group for existing VMs.

## **Can I modify VMs that are replicating by adding or resizing disks?**

For VMware replication to Azure, you can modify the disk size of source VMs. To add new disks, you must add the disk and reenable protection for the VM.

## **Can I migrate on-premises machines to a new vCenter Server without impacting ongoing replication?**

Refer to the [guidance](#) to migrate machines to a new vCenter.

## **Can I replicate to a cache or target storage account that has a virtual network (with Azure Firewalls) configured on it?**

No, Site Recovery doesn't support replication to Azure Storage on virtual networks.

## **What is the frequency of generation of crash-consistent recovery points?**

Site Recovery generates crash-consistent recovery points every five minutes.

## Can I change an already replicating machine from one to another Recovery Services vault?

Switching Recovery Services vaults, when the replication is already ongoing, isn't supported. To do so, replication needs to be disabled and enabled again. Additionally, the mobility service agent, installed on the source machine, must be un-configured so that it can be configured to a new vault. Use the following commands to perform the unregistration:

*For Windows machines:*

```
C:\Program Files (x86)\Microsoft Azure Site  
Recovery\agent\UnifiedAgentConfigurator.exe /Unconfigure true
```

*For Linux machines:*

```
/usr/local/ASR/Vx/bin/UnifiedAgentConfigurator.sh -q -U true -c CSPrime
```

## Can I change an already replicating machine from one region to another region?

Switching region, when the replication is already ongoing isn't supported. To do so, replication needs to be disabled and enabled again. Additionally, to avoid reboot there is no need to uninstall the mobility agent, mobility agent installed on the source machine, must be un-configured using the commands mentioned. Use the following commands to perform the unregistration:

*For Windows machines:*

```
C:\Program Files (x86)\Microsoft Azure Site  
Recovery\agent\UnifiedAgentConfigurator.exe /Unconfigure true
```

*For Linux machines:*

```
/usr/local/ASR/Vx/bin/UnifiedAgentConfigurator.sh -q -U true -c CSPrime
```

## Component upgrade

## My version of the Mobility services agent or configuration server is old, and my upgrade failed. What do I do?

Site Recovery follows the N-4 support model. [Learn more](#) about how to upgrade from old versions.

## Where can I find the release notes and update rollups for Azure Site Recovery?

[Learn about new updates](#), and get rollup information.

## Where can I find upgrade information for disaster recovery to Azure?

[Learn about upgrading](#).

## Do I need to reboot source machines for each upgrade?

Rebooting is recommended but not mandatory for each upgrade. [Learn more](#).

## Configuration server

### What does the configuration server do?

The configuration server runs the on-premises Site Recovery components, including:

- The configuration server itself. The server coordinates communications between on-premises components and Azure, and manages data replication.
- The process server, which acts as a replication gateway. This server:
  1. Receives replication data.
  2. Optimizes the data with caching, compression, and encryption.
  3. Sends the data to Azure Storage. The process server also does a push install of the Mobility Service on VMs and performs automatic discovery of on-premises VMware VMs.

- The master target server, which handles replication data during failback from Azure.

[Learn more](#) about the configuration server components and processes.

## Where do I set up the configuration server?

You need a single, highly available, on-premises VMware VM for the configuration server. For physical server disaster recovery, install the configuration server on a physical machine.

## What do I need for the configuration server?

Review the [prerequisites](#).

## Can I manually set up the configuration server instead of using a template?

[Create the configuration server VM](#) by using the latest version of the Open Virtualization Format (OVF) template. If you can't use the template (for example, if you don't have access to the VMware server), [download](#) the setup file from the portal and set up the configuration server.

## Can a configuration server replicate to more than one region?

No. To replicate to more than one region, you need a configuration server in each region.

## Can I host a configuration server in Azure?

Although it's possible, the Azure VM running the configuration server needs to communicate with your on-premises VMware infrastructure and VMs. This communication adds latency and impacts ongoing replication.

## How do I update the configuration server?

[Learn](#) how to update the configuration server.

- Find the latest update information on the [Azure updates page](#) .
- Download the latest version from the portal. Or, download the latest version of the configuration server directly from the [Microsoft Download Center](#) .

- If your version is more than four versions older than the current version, see the [support statement](#) for upgrade guidance.

## Should I back up the configuration server?

Take regular scheduled backups of the configuration server.

- For successful failback, the VM you're failing back must exist in the configuration server database.
- The configuration server must be running and in a connected state.
- [Learn more](#) about common configuration server management tasks.

## When I set up the configuration server, can I download and install MySQL manually?

Yes. Download MySQL and place it in the C:\Temp\ASRSetup folder. Then, install it manually.

When you set up the configuration server VM and accept the terms, MySQL appears as **Already installed** in **Download and install**.

## Can I avoid downloading MySQL but let Site Recovery install it?

Yes. Download the MySQL installer and place it in the C:\Temp\ASRSetup folder. When you set up the configuration server VM, accept the terms and select **Download and install**. The portal uses the installer that you added to install MySQL.

## Can I use the configuration server VM for anything else?

No. Use the VM only for the configuration server.

## Can I clone a configuration server and use it for orchestration?

No. Set up a fresh configuration server to avoid registration issues.

## Can I change the vault in which the configuration server is registered?

No. After a vault is associated with the configuration server, you can't change it. [Learn](#) about registering a configuration server with a different vault.

## Can I use the same configuration server for disaster recovery of both VMware VMs and physical servers?

Yes, but note that a physical machine can fail back only to a VMware VM.

## Where can I download the passphrase for the configuration server?

[Learn](#) how to download the passphrase.

## Where can I download vault registration keys?

In the Recovery Services vault, select **Configuration Servers** under **Site Recovery Infrastructure > Manage**. Then, in **Servers**, select **Download registration key** to download the vault credentials file.

## Can a single configuration server be used to protect multiple vCenter instances?

Yes, a single configuration server can protect VMs across multiple vCenters. There's no limit on how many vCenter instances you can add to the configuration server. However, the limits for how many VMs a single configuration server can protect do apply.

## Can a single configuration server protect multiple clusters within vCenter?

Yes, Azure Site Recovery can protect VMs across different clusters.

## Process server

### How can I ensure high availability of the process server?

By configuring more than one process server, you gain the flexibility to move protected machines from an unhealthy process server to a working process server. You must explicitly and manually

initiate the movement of a machine from one process server to another. For more information, see [moving VMs between process servers](#).

## Failover and failback

### Can I use the on-premises process server for failback?

Use a process server in Azure for failback to avoid data transfer latencies. Additionally, if you separate the source VMs network from the Azure-facing network in the configuration server, you must use the process server created in Azure for failback.

### Can I keep the IP address on failover?

Yes, you can keep the IP address on failover. Ensure that you specify the target IP address in the **Compute and Network** settings for the VM before failover. Also, shut down machines at the time of failover to avoid IP address conflicts during failback.

### Can I change the target VM size or VM type before failover?

Yes, you can change the type or size of the VM at any time before failover. In the portal, use the **Compute and Network** settings for the replicated VM.

### How far back can I recover?

For VMware to Azure, the oldest recovery point you can use is 15 days.

### How does the pruning of recovery points happen?

Crash-consistent recovery points are generated in every five minutes. App-consistent snapshots are generated based on the input frequency entered by you. Beyond two hours, pruning of recovery points may happen based on the retention period that you input. Following are the scenarios:

 Expand table

| Retention Period input | Pruning mechanism                                                                                                                                                                                                                                                             |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0 day                  | No recovery point saved. You can failover only to the latest point.                                                                                                                                                                                                           |
| 1 day                  | One recovery point saved per hour beyond the last two hours.                                                                                                                                                                                                                  |
| 2 - 7 days             | One recovery point saved per two hours beyond the last two hours.                                                                                                                                                                                                             |
| 8 - 15 days            | One recovery point saved per two hours beyond last two hours for 7 days. After that, one recovery point saved per four hours.<br>App-consistent snapshots are also pruned based on the duration mentioned above even if you input a lesser app-consistent snapshot frequency. |

## Do increases in recovery point retention increase storage costs?

Yes. For example, if you increase retention from one day to three days, Site Recovery saves recovery points for an extra two days. The added time incurs storage charges. Earlier, the system saved recovery points per hour for one day. Now, it saves recovery points per two hours for three days. For more information, see [pruning of recovery points](#). So the system saves 12 more recovery points. As an example only, if a single recovery point had delta changes of 10 GB, with a per-GB cost of \$0.16 per month, the additional charges are  $\$1.60 \times 12$  per month.

## How do I access Azure VMs after failover?

After failover, you can access Azure VMs over a secure internet connection, over a site-to-site VPN, or over Azure ExpressRoute. To connect, you must prepare several things. [Learn more](#).

## Is failed-over data resilient?

Azure is designed for resilience. Site Recovery is engineered for failover to a secondary Azure datacenter, as required by the Azure service-level agreement (SLA). When failover occurs, your metadata and vaults stay in the same geographic region that you chose for your vault.

## Is failover automatic?

[Failover](#) isn't automatic. You start a failover by making a single selection in the portal, or you can use [PowerShell](#) to trigger a failover.

## Can I fail back to a different location?

Yes. If you fail over to Azure, you can fail back to a different location if the original one isn't available. [Learn more.](#)

## Why do I need a VPN or ExpressRoute with private peering to fail back?

When you fail back from Azure, data from Azure is copied back to your on-premises VM, and private access is required.

## Can I use On-demand capacity reservation with VMware(also Physical) to IaaS scenario?

No. Azure Site Recovery can't use On-demand capacity reservation unless it's Azure to Azure scenario.

## The application license is based on UUID of VMware virtual machine. Is the UUID of a VMware virtual machine changed when it is failed over to Azure?

Yes, the UUID of the Azure virtual machine is different from the on-premises VMware virtual machine. However, most application vendors support transferring the license to a new UUID. If the application supports it, the customer can work with the vendor to transfer the license to the VM with the new UUID.

## Automation and scripting

### Can I set up replication with scripting?

Yes. You can automate Site Recovery workflows by using the REST API, PowerShell, or the Azure SDK. [Learn more.](#)

## Performance and capacity

### Can I throttle replication bandwidth?

Yes. [Learn more](#).

## Next steps

- [Review](#) support requirements.
- [Set up](#) VMware to Azure replication.

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